

Python Final Project

Requirements, Scoring, and Example Topics

1. Project Goal

The goal of the project is to **independently** create your own standalone application in Python that implements a chosen topic and uses the language elements discussed during classes. You may choose a project from the list of example topics or propose your own, provided it meets all technical requirements.

2. Technical Requirements and Scoring (max. 40 pts)

2.1. Program Structure and Architecture (8 pts)

- sensible division into modules and files — **3 pts**
- correct use of classes — **3 pts**
- readability and organization of code — **2 pts**

2.2. Python Language Elements (20 pts)

Each element must be used naturally and with justification. All elements together must form a coherent project. There should not be any code fragments unrelated to the logic of the final application.

- control statements (**if**, **for**, **while**) — **1 pt**
- operators (arithmetic, logical, comparison) — **1 pt**
- functions + at least one **lambda** — **3 pts**
- custom decorator — **3 pts**
- collections (lists, dictionaries, sets) — **2 pts**
- comprehensions (list/dict/set comprehensions) — **2 pts**
- generator (generator function or generator expression) — **2 pts**
- file handling + context manager (**with**) — **3 pts**
- serialization (JSON, CSV, or pickle) — **2 pts**
- regular expressions — **1 pt**

2.3. Error and Exception Handling (4 pts)

- meaningful error messages — **2 pts**
- at least one custom exception — **2 pts**

2.4. User Interface (4 pts)

- clear menu/commands — **2 pts**
- predictable program behavior — **2 pts**

2.5. Documentation (4 pts)

- README file with description — **2 pts**
- run instructions — **2 pts**

2.6. Quality and Creativity (4 pts)

- code aesthetics, extensions, thoughtful logic — **4 pts**

3. Grading Rubric

Criterion	Description	Points
Program structure	Modules, classes, readability	0–8
Control statements	Correct usage	0–1
Operators	Logical/comparison operators	0–1
Functions and lambdas	Helper functions, lambda	0–3
Decorator	Custom decorator	0–3
Collections	Lists, dictionaries, sets	0–2
Comprehensions	List/dict/set comprehensions	0–2
Generator	Generator or generator expression	0–2
Files + with	File handling	0–3
Serialization	JSON/CSV/pickle	0–2
Regex	Meaningful usage	0–1
Exceptions	Error handling + custom exception	0–4
User interface	Menu, messages	0–4
Documentation	README, comments	0–4
Creativity	Aesthetics, extensions	0–4

Additional Points

It is possible to earn additional points for extending the project beyond the listed requirements. These points count toward activity points, of which you can earn a maximum of 10 during the entire semester.

4. Project Submission

- The project must be submitted through the appropriate Teams assignment as a .zip file named *sXXXXX_projekt.zip*.
- The zip file may contain only text files such as *.py*, *.md*, etc. This means that *.venv*, *.idea*, and *.git* must **not** be included.
- Submitting the project in an incorrect format will result in a **5-point deduction** from the final project score.

5. Example Project Topics

5.1. Practical Projects

- Habit Tracker — tracking habits, statistics, JSON storage.
- Mini Library — books, users, loans, regex-based search.
- Invoicing System — clients, items, NIP validation via regex.
- Text Analyzer — word statistics, pattern searching.
- Bill Analyzer — bill analysis, anomaly detection.

5.2. Utility Projects

- Log Parser — log analysis, JSON reports.
- File Cleaner — duplicate detection, suspicious filenames.
- Unit Converter — regex command parser, plugins.

5.3. Creative Projects

- Text-based RPG — player, enemies, event generator.
- Sudoku Solver — validation, solving, saving results.
- Quiz Master — creating and solving quizzes.

6. Defense

- You must attend the defense at the date specified by the instructor. Failure to attend results in **0 points for the project**.
- If the project does not start during the defense, **10 points will be deducted**.
- If the project starts but an unhandled error occurs during the presentation, **5 points will be deducted**.
- During the defense, the instructor may ask 3 to 5 questions to verify whether the student knows and understands their code.

- If the student does not answer at least 3 questions correctly, this results in **0 points for the project**.

7. Summary

Most Important Information

- Maximum: **40 points**.
- The project must include: functions, lambdas, decorator, classes, collections, comprehensions, generator, regex, files + **with**, serialization, exceptions.
- A user interface is required (CLI or simple UI).
- Include a README file with description and run instructions.
- You may choose a topic from the list or propose your own.
- Incorrect submission format results in a 5-point deduction.
- During the defense, you may lose points if your project fails to start, throws errors, or if you cannot answer at least 3 instructor questions.